

Vigenère cipher cracker

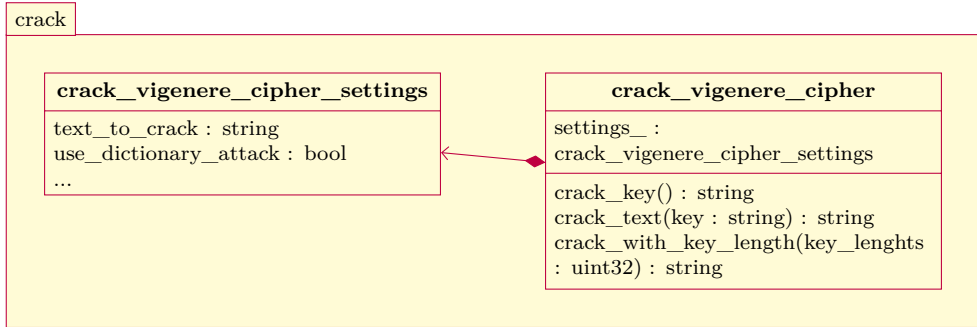
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Class program structure



Basic code structure class-vice

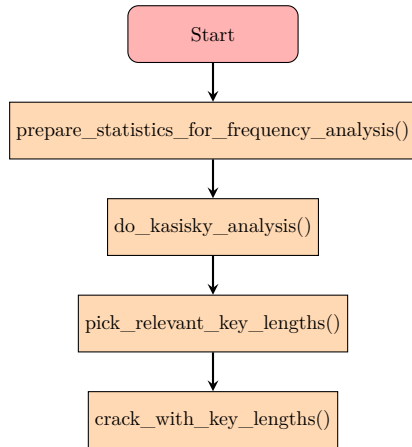


crackVigenereCipher function

```
1 function [crackedKey, crackedText] = crackVigenereCipher(cipherText)
2     settings = crack.generate_crack_vigenere_cipher_settings(
3         cipherText);
4     cracker = crack.crack_vigenere_cipher(settings);
5     crackedKey = cracker.crack_key();
6     crackedText = cracker.crack_text(crackedKey);
7 end
```

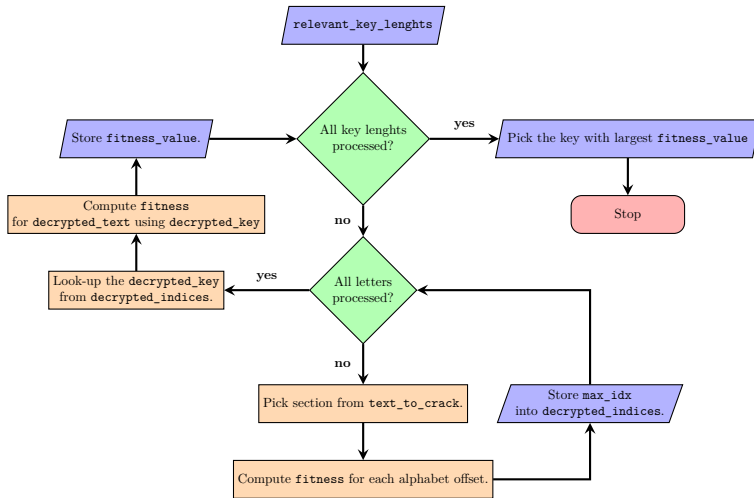


crack.crack_key() flowchart





crack.crack_with_key_length() flowchart



Fitness computation



- Suppose two sequences $A = (a_n)_{n=1}^N$ and $B = (b_n)_{n=1}^N$, $a_n, b_n \in \mathbb{N}$. Then we define a fitness operator

$$\mathcal{F}(A, B) := \frac{\langle A | B \rangle_{\ell^2}}{\|A\|_{\ell^2} \|B\|_{\ell^2}} = \frac{\sum_{n=1}^N a_n b_n}{\sqrt{\sum_{n=1}^N a_n^2} \sqrt{\sum_{n=1}^N b_n^2}}.$$

Questions?